Missing Mass and Minimum Wages: Distributional Effects of Three Minimum Wage Increases in Peru

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Abstract

I study the distributional effects of three minimum wage (MW) increases in Peru—from S/750 to S/1,025 between 2016 and 2022—using a pre-post distributional estimator adapted from Cengiz et al. (2019) for a national MW setting without control jurisdictions. Bunching ratios range from 0.70 to 0.83: for every 100 jobs displaced below the new minimum, 70–83 reappear above it; placebo tests at artificial thresholds produce ratios 7 times smaller, confirming the signal is MW-specific. Extending the excess window to S/500 raises R to 0.94 (Event C), indicating workers disperse broadly above the new floor. The formal missing mass is accounted for in part by a 15–21 percentage point increase in the self-employment share within the affected wage range across all three events, with the absolute count of self-employed workers rising in Events A

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(+8.7%) and B (+5.1%); Event C shows a share increase (+15.3pp) but a slight absolute decline, consistent with post-COVID re-formalization dynamics. Together these patterns are consistent with transitions to informal self-employment rather than job destruction. Results are replicated on EPE Lima quarterly data (ratios 0.73–1.03) using independent formality definitions and tighter 6-month event windows. Employment effects cannot be identified through regression-based methods: event-study pre-trends fail for Events B and C, the Kaitz instrumental variable is weak ($F < 3$), and panel attrition exceeds 76%. Hours show no adjustment ($\leq \pm 1.1$h/week); wage compression of 3–7 percentage points is present but 41–92% is mechanical composition. The dominant effect of Peru's MW increases within the studied Kaitz range (0.52–0.57) is redistribution within the formal wage distribution combined with institutional reallocation from formal employment to informal self-employment, not measurable job destruction.

JEL Codes: J31, J38, O17

Keywords: minimum wage, bunching estimator, wage distribution, Peru, informality, self-employment

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1 Introduction

Peru raised its minimum wage (*Remuneración Mínima Vital*, RMV) three times between 2016 and 2022, from S/750 to S/1,025—a cumulative 37% nominal increase. Throughout this period, formal employment remained stable and the formality rate recovered from its COVID-19 dip. What happened to the wage distribution? Did workers near the minimum wage floor receive raises, or did they lose their jobs?

This paper uses a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for Peru's national MW setting, to answer these questions. The estimator compares the formal wage distribution before and after each MW increase, measuring "missing mass" below the old threshold and "excess mass" at the new one. The ratio of excess to missing mass—the bunching ratio—indicates what fraction of displaced jobs reappear above the new floor.

The key challenge is that Peru has a single national MW with no sub-national variation in the policy. Cengiz et al. (2019) construct their counterfactual by comparing U.S. states with MW changes to states without—a difference-in-differences design. This is not feasible in Peru. Instead, I follow the methodological precedent of Harasztosi and Lindner (2019), who analyze Hungary's national MW using the pre-reform wage distribution as the counterfactual, with a background correction from the upper tail of the distribution.

I analyze three MW events: Event A (S/750→S/850, May 2016), Event B (S/850→S/930, April 2018), and Event C (S/930→S/1,025, May 2022). Bunching ratios range from 0.70 to 0.83: for every 100 displaced jobs, 70–83 reappear above the new floor, rising to 94 at wider excess windows. The formal missing mass is partially accounted for by a 15–21 percentage point increase in self-employment within the same wage range. Placebo tests confirm the signal is MW-specific (7× larger than at artificial thresholds), and all three regression-based employment identification strategies fail. This paper extends previous work on Peru's labor market (Jaramillo Baanante, 2004; Céspedes, 2006; Céspedes and Sánchez, 2014; Jaramillo Baanante, 2012; Chávez, 2024, 2023) by providing a distributional estimator

with a positive answer to where the missing workers go—institutional reallocation to informal self-employment, not job destruction.

This paper makes three contributions. First, it provides the first distributional analysis of Peru's minimum wage using three identifiable events across 2015–2023, documenting redistribution ratios of 0.70–0.83 and a 15–21 percentage point increase in self-employment in the affected wage range, together indicating institutional reallocation from formal employment to informal self-employment rather than job destruction (contributing to Harasztosi and Lindner, 2019; Saltiel and Urzúa, 2018; Gindling and Terrell, 2021; Engbom and Moser, 2022; Corcuera, 2024). Second, it systematically documents why regression-based identification strategies fail in national-MW settings: three approaches—event-study DiD, Kaitz IV, and panel decomposition—all fail for structural reasons, offering a methodological cautionary note for researchers working in similar institutional contexts (Card and Krueger, 1994; Neumark and Shirley, 2022; Manning, 2021; Dube, 2019; Belman and Wolfson, 2014). Third, it identifies a non-obvious but critical requirement for discrete-bin distributional estimators: bin boundaries must align with the policy threshold. Misalignment at S/50 bins causes the bunching ratio to collapse by 75%, an artifact that could lead researchers using coarser wage bins to conclude large-scale job destruction where there is none.

2 Institutional Context

2.1 Peru's Minimum Wage System

The *Remuneración Mínima Vital* (RMV) is set by executive decree and applies to all registered formal dependent workers (*empleados* and *obreros*). It is not indexed to inflation or productivity and changes only when the government issues a new decree. Enforcement is conducted by SUNAFIL (*Superintendencia Nacional de Fiscalización Laboral*), with inspections concentrated in Lima and large firms.

Peru's labor market is characterized by high informality: approximately 72–75% of em-

ployed workers are informal by INEI's comprehensive definition (Yamada, 1996; Chacaltana, 2001; Jaramillo and López, 2006; Díaz, 2014). The RMV has been studied as a labor market anchor and its effects on wages, employment, and formality have been documented across multiple reform episodes (Jaramillo Baanante, 2004; Céspedes, 2006; Céspedes and Sánchez, 2014; Jaramillo Baanante, 2012; Corcuera, 2024). Previous studies of Peru's MW include Jaramillo Baanante (2004), who finds effects concentrated at the bottom of the distribution with little support for lighthouse effects; Céspedes (2006) and Céspedes and Sánchez (2014), who estimate employment elasticities using EPE Lima panel data; Jaramillo Baanante (2012), who finds MW increases raise employment for some informal workers; and Corcuera (2024), who documents reallocation to informal self-employment. Yamada (1996) provides the foundational analysis of Peru's informal sector structure. The MW directly affects roughly 10–15% of the workforce classified as formal dependent workers under INEI's formality variable (`ocupinf == 2`), which requires social security registration or a written employment contract.

2.2 The Three MW Events

Table 1 summarizes the three MW changes analyzed in this paper. Each event is separated by at least two years, allowing clean pre-post comparisons using annual ENAHO surveys.

Table 1: Minimum Wage Events Analyzed

Event	Pre-MW	Post-MW	Change	Effective	Analysis years
A	S/750	S/850	+13.3%	May 2016	Pre: 2015, Post: 2017
B	S/850	S/930	+9.4%	April 2018	Pre: 2017, Post: 2019
C	S/930	S/1,025	+10.2%	May 2022	Pre: 2021, Post: 2023

Note: No 2020 survey due to COVID-19. Event C's pre-period (2021) reflects post-pandemic labor market conditions.

2.3 Department-Level Kaitz Variation

The Kaitz index (MW divided by the weighted-median formal wage by department) ranges from 0.39 to 1.09 across departments and years. Ica (department 11) has the highest Kaitz (0.94 in 2015), driven by formal agricultural workers on agro-export farms earning near-minimum monthly wages despite full-time hours (median 48 hours/week). Huancavelica (department 09) has a low Kaitz (≈ 0.50) because approximately 85% of its formal workers are public-sector employees earning above the MW.

This variation appears to primarily reflect differences in economic structure—the shares of agriculture and public-sector employment—rather than differential MW exposure. It does not generate sufficient first-stage power for IV identification ($F < 3$ in all events; see Appendix C), and employment pre-trends fail for Events B and C (Appendix B). The Kaitz index is used in this paper as a descriptive measure of MW bindingness, not as an instrument.

3 Data

3.1 ENAHO Module 500

The primary data source is the *Encuesta Nacional de Hogares* (ENAHO) Module 500 (*Empleo e Ingresos*), conducted annually by INEI. I use cross-sectional data from eight years: 2015–2019, 2021–2023 (no 2020 survey due to COVID-19).

The analysis sample consists of formal dependent workers with monthly wages between S/1 and S/6,000 and positive survey weights. The filter chain is: employed (`ocu500 == 1`), dependent worker (`p507` $\in \{3, 4, 6\}$: *empleado*, *obrero*, *trabajador del hogar*), formal (`ocupinf == 2`), wage $\in (0, 6000)$, weight > 0. The wage variable is `p524a1` (monthly cash income from primary occupation). The weight variable is `fac500a`. Department is identified from the first two digits of `ubigeo`.

3.2 Summary Statistics

Table 2 presents summary statistics for the analysis sample across all eight years.

Table 2: Summary Statistics — Formal Dependent Workers, ENAHO 2015–2023

	2015	2016	2017	2018	2019	2021	2022	2023
MW (S/.)	750	850	850	930	930	930	1,025	1,025
N	10,059	11,488	10,895	11,723	10,922	8,946	9,554	9,838
Median wage	1,400	1,500	1,500	1,673	1,700	1,800	1,800	1,863
% at MW	4.6	4.8	6.0	4.8	6.5	7.8	5.7	8.2
% below MW	10.9	14.0	10.4	14.6	12.5	11.5	16.9	12.6
Median Kaitz	0.536	0.567	0.567	0.556	0.547	0.517	0.569	0.550
Mean hrs/wk	46.4	46.2	46.4	46.3	46.2	46.8	46.6	46.5
Formality (%)	26.7	27.9	27.4	27.6	27.3	23.2	24.3	26.1

Note: N = unweighted formal dependent workers with monthly wage S/1–6,000 and positive survey weight. % at MW = share earning MW±25 (one bin). % below MW = share earning below MW. Median Kaitz = MW / median wage, national. Formality rate = formal dependent / all employed. The 2020 ENAHO is excluded due to reduced sample size and methodological modifications during COVID-19, which limit comparability with other years. COVID-19 year 2021 shows depressed formality reflecting pandemic-period job losses.

Median wages rise from S/1,400 (2015) to S/1,863 (2023), a 33% nominal increase that roughly tracks the 37% MW increase. The Kaitz index ranges from 0.517 to 0.569 with no clear trend; it does not monotonically increase over time. The share below MW rises sharply in pre-event years (14.0% in 2016, 14.6% in 2018, 16.9% in 2022) and then falls as wages adjust to the new floor.[1] Weekly hours are stable at 46–47 hours throughout the period.

[1]The elevated below-MW shares in pre-event years reflect gradual erosion of the real MW between increases, not anticipation effects. These workers are the treatment population: they earn below the new MW and are directly affected by the increase. Their presence in the pre-period is a feature of the identification,

4 Empirical Strategy

4.1 Pre-Post Distributional Estimator

I use a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for a national MW setting without control jurisdictions. Cengiz et al. (2019) construct the counterfactual using U.S. states that did not raise their MW. Peru has a single national MW; I use the pre-period wage distribution with a background correction from the upper tail ($> 2 \times \text{MW}_{\text{new}}$) instead. The closer methodological precedent is Harasztosi and Lindner (2019), who study Hungary's national MW.

The wage distribution is discretized into S/25 bins over the range S/0–S/6,000, weighted by survey expansion factors (`fac500a`). For each bin b, I compute the share of the workforce in the pre-period (s_b^{pre}) and post-period (s_b^{post}), and compute the raw delta: $\Delta_b = s_b^{\text{post}} - s_b^{\text{pre}}$.

The counterfactual correction uses an inverse-absolute-weighted average of Δ_b in the "clean zone" (bins with centers above $2 \times \text{MW}_{\text{new}}$), where the MW is unlikely to affect the distribution:

$$\bar{\Delta}_{\text{clean}} = \frac{\sum_{b \in \mathcal{C}} w_b \cdot \Delta_b}{\sum_{b \in \mathcal{C}} w_b}, \quad w_b = \frac{1}{|\Delta_b| + \epsilon}$$

where $\epsilon = 10^{-8}$. The adjusted delta is $\Delta_b^{\text{adj}} = \Delta_b - \bar{\Delta}_{\text{clean}}$.

Bins with $|\Delta_b| \approx 0$ receive higher weight because they represent wage regions where the pre-to-post change is minimal, providing the cleanest estimate of the background trend. The weighting prevents a small number of bins with anomalously large movements—which may reflect sector shocks rather than MW effects—from dominating the correction. Table 5 shows results are robust to uniform weighting and alternative clean zone thresholds.

not a threat—the bunching estimator measures precisely how this below-MW mass redistributes after the policy change.

Missing mass (negative-only bins in the affected zone):

$$B^{\text{miss}} = \sum_{\substack{b \in [0.85 \times \text{MW}_{\text{old}},\ \text{MW}_{\text{new}}) \\ \Delta_b^{\text{adj}} < 0}} \left| \Delta_b^{\text{adj}} \right| \tag{1}$$

Excess mass (positive-only bins in the excess zone):

$$B^{\text{exc}} = \sum_{\substack{b \in [\text{MW}_{\text{new}},\ \text{MW}_{\text{new}} + 250) \\ \Delta_b^{\text{adj}} > 0}} \Delta_b^{\text{adj}} \tag{2}$$

The **bunching ratio** is $R = B^{\text{exc}}/B^{\text{miss}}$. A ratio of 1 indicates perfect redistribution: all displaced workers reappear above the new MW. A ratio below 1 indicates that some fraction of the missing mass does not reappear, consistent with job loss, informalization, or transitions beyond the S/250 excess window.

This estimator does not require between-unit parallel trends and is valid for all three events. It does assume that the upper tail of the distribution ($> 2 \times \text{MW}_{\text{new}}$) provides a valid counterfactual for background wage growth in the MW-relevant range. This assumption could be violated if high-wage and low-wage formal employment have different cyclical dynamics—for example, if recessions disproportionately contract low-wage formal employment. The placebo tests in Section 5 provide a within-sample check: if the upper-tail correction were systematically biased, false-positive bunching signals would appear at artificial thresholds.

Bin-width alignment. An underappreciated requirement of discrete-bin distributional estimators is that bin boundaries must align with the MW threshold: the MW value must be divisible by the chosen bin width. When this condition fails, the bunching spike at MW_{new} straddles a bin boundary, and workers pushed exactly to the new floor are classified as "missing" rather than excess, causing catastrophic downward bias in R. At S/50 bins, for example, $\text{MW}_{\text{new}} = 930$ falls in bin $[900, 950)$ with center $925 < 930$; workers earning S/930–

S/949 are counted in the *missing* zone rather than the excess zone, collapsing R from 0.829 to 0.206—a 75% downward bias indistinguishable from genuine employment destruction. The S/25 canonical grid is the finest resolution at which both Event A ($\text{MW}_{\text{new}} = 850$) and Event C ($\text{MW}_{\text{new}} = 1{,}025$) are aligned. We recommend that future applications of bunching methods verify bin-boundary alignment before reporting results. Table 6 reports sensitivity across six bin widths; R is stable at aligned widths (range of 0.009 across S/10–S/25 for Event B) and collapses at severely misaligned widths.

4.2 Bootstrap Inference

Standard errors are computed by block-bootstrapping individuals within each pre/post cross-section (1,000 replications). For each bootstrap draw, I resample with replacement and recompute the full bunching ratio. The 95% confidence interval is the 2.5th and 97.5th percentiles of the bootstrap distribution.

4.3 Hours DiD

The intensive-margin hours response is measured by a 2×2 DiD of weighted mean weekly hours (`p513t`). The treatment group is workers in the affected wage zone $[0.85\times\text{MW}_{\text{old}}, \text{MW}_{\text{new}})$ in the pre-period and $[\text{MW}_{\text{new}}, 1.3 \times \text{MW}_{\text{new}}]$ in the post-period (tracking where displaced workers land). The control group is workers in $[1.5 \times \text{MW}_{\text{new}}, 3.0 \times \text{MW}_{\text{new}}]$ in both periods.

4.4 Failed Identification Strategies (Appendices B–C)

I attempted three regression-based identification strategies for employment effects, all of which fail. The event-study DiD uses department-level Kaitz variation; pre-trends are violated for Events B ($p = 0.007$) and C ($p = 0.017$). The Kaitz IV produces F-statistics of 1.5, 2.6, and 0.1—all far below the Stock-Yogo threshold. See Appendices B and C for details.

5 Bunching Results

5.1 Main Estimates

Table 3 presents the bunching estimates for all three events and three populations.

Table 3: Pre-Post Bunching Estimates — ENAHO Cross-Section

Event	Group	Missing (pp)	Excess (pp)	Ratio
A (750→850)	Formal dep	6.78	4.72	0.696
	Informal dep	4.45	5.42	1.218
	All workers	5.41	4.72	0.872
B (850→930)	Formal dep	8.03	6.66	0.829
	Informal dep	5.28	5.21	0.987
	All workers	6.44	5.81	0.901
C (930→1,025)	Formal dep	13.02	10.80	0.830
	Informal dep	5.50	4.46	0.811
	All workers	8.29	6.84	0.824

Note: Missing (pp) = share lost from $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ using negative-only bins. Excess (pp) = share gained at $[\mathrm{MW}_{\mathrm{new}}, \mathrm{MW}_{\mathrm{new}} + 250)$ using positive-only bins. Background correction: inverse-abs-weighted mean from clean zone ($> 2 \times \mathrm{MW}_{\mathrm{new}}$). Bins: S/25, range S/0–6,000.

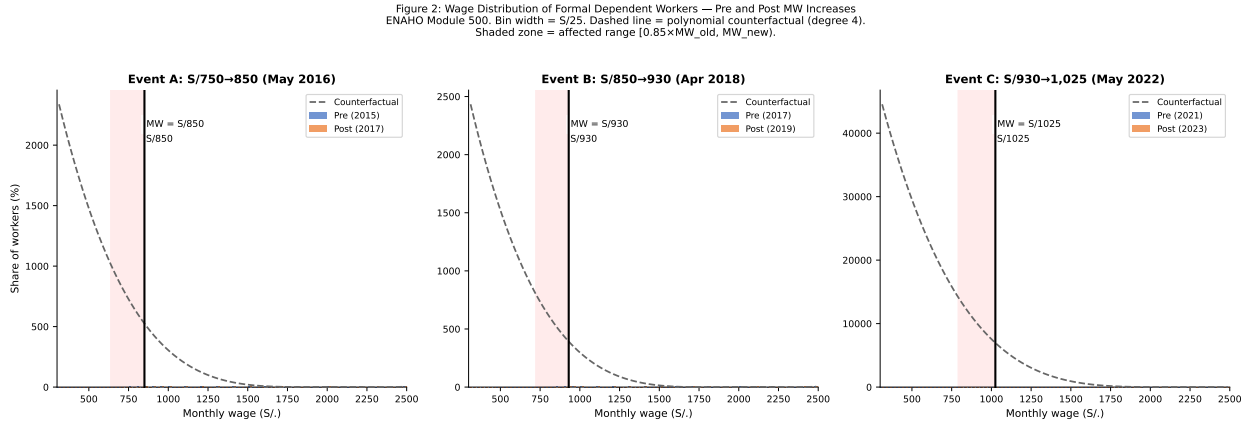

Figure 1: Wage Distributions Before and After MW Increases

Note: Three panels show the formal dependent wage distribution (S/25 bins, weighted by `fac500a`) before (blue) and after (orange) each MW increase. Vertical lines mark the old (dashed gray) and new (solid red) MW. Shaded region = affected zone $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$. Y-axes show workforce share (percentage points); scales are common across panels to facilitate cross-event comparison.

Formal dependent workers show ratios consistently below 1 across all three events (0.696–0.830), indicating that more workers leave the affected zone than reappear at the new MW. Missing mass increases from 6.78pp (Event A) to 13.02pp (Event C), reflecting the growing MW bite as the RMV rises relative to the wage distribution.

Table 4 reports the unweighted observation counts in the affected and excess zones, confirming that the bunching signal is not driven by sparse cells. The affected zone contains 829–1,163 observations pre-policy; the excess zone contains 1,566–1,657 observations post-policy.

Table 4: Effective Sample Counts in Affected and Excess Zones

Event	N_{affected} (pre) $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$	N_{excess} (post) $[\text{MW}_{\text{new}}, \text{MW}_{\text{new}} + 250)$	N_{total} (pre)
A (750→850)	829	1,657	10,195
B (850→930)	1,032	1,566	11,090
C (930→1,025)	1,163	1,572	9,175

Note: Formal dependent workers. Unweighted counts. Affected zone = directly compressed zone below the new MW. Excess zone = first S/250 above the new MW (canonical excess window). $N_{\text{excess}} > N_{\text{affected}}$ because the excess zone captures both workers pushed up from the affected zone and new entrants; some excess-zone workers may have always been above the affected zone.

Bootstrap 95% confidence intervals are $[0.567, 0.896]$ (A), $[0.716, 1.016]$ (B), and $[0.716, 0.960]$ (C). Events A and C reject $R = 1$ at the 95% level; Event B does not—approximately 4.7% of bootstrap draws produce $R \geq 1$. For Event B, the data are consistent with both full redistribution and displacement of up to 28% of affected workers.

Informal workers show a ratio above 1 in Event A (1.218), approximately 1 in Event B (0.987), and below 1 in Event C (0.811). There is no consistent spillover pattern.

Informal lighthouse effect. Event A's informal bunching ratio of 1.218—excess mass exceeding missing mass at the formal MW threshold—is consistent with a lighthouse effect whereby the formal MW serves as a reference point for informal wage-setting (Jaramillo Baanante, 2004). If informal employers use the MW as a focal point for wage offers, an MW increase would shift informal wages upward even without legal enforcement, generating excess mass at the new floor. This pattern weakens across events ($1.218 \rightarrow 0.987 \rightarrow 0.811$), suggesting the lighthouse effect operates at lower MW levels but fades as the MW rises relative to informal productivity. The welfare implication is nuanced: if informal workers receive MW-indexed

raises via the lighthouse mechanism, the self-employment transition documented below may preserve income levels even without formal social security coverage.

Where do the missing workers go? A natural concern is that the missing formal mass simply re-emerges as informal employment near the old MW floor. This is not supported by the data. The informal wage share in the affected zone $[\mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ *decreases* post-reform in all three events: by 4.4 percentage points in Event A, 4.4pp in Event B, and 5.3pp in Event C. If formal workers were simply reclassified as informal, the informal share in this zone would rise. Instead, both formal *and* informal employment in the affected zone contract together, suggesting the missing workers move to self-employment, reduce hours below the monthly equivalent of the MW, or exit formal labor force participation entirely. Cross-sectional designs cannot trace individuals, so this interpretation is necessarily indirect. At the department level, higher Kaitz indices are not associated with differential changes in the formality rate (OLS coefficient: $\hat{\beta} = +0.020$, $p = 0.225$, $n = 24$ departments, Event A), confirming no measurable formal-to-informal transition even in high-exposure departments. These patterns indicate the formal missing mass does not transition into informal dependent employment; instead, workers exit dependent employment arrangements altogether.

Supplementing this analysis with self-employment income data (ENAHO `p530a`, monthly net business income) reveals a strong self-employment signature in the affected zone. Among all workers reporting income in $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$—combining dependent and self-employed workers—the self-employed share rises from 34–38% pre-reform to 49–57% post-reform, a jump of 15–21 percentage points across all three events. Crucially, the *absolute count* of self-employed workers in the zone increases in Events A (+8.7%: 496,000 → 539,000) and B (+5.1%: 504,000 → 529,000), while formal and informal dependent employment in the same zone contracts sharply. This is the strongest direct evidence that part of the formal missing mass transitions to self-employment at similar income levels rather than to non-employment. This finding is corroborated by Corcuera (2024), who identifies a similar reallocation mechanism using an independent empirical strategy.

Event C is the exception: while the self-employment share in the affected zone rises (+15.3pp), the absolute count of self-employed workers declines slightly (−3.5%: 596,000 → 574,000). This likely reflects the post-COVID recovery pulling workers back into formal employment between 2021 and 2023—the formal workforce grew 12.5% over this period, and some workers who would otherwise have remained in self-employment re-entered formal arrangements. Event C's formal missing mass (13.02pp) is therefore partially absorbed by the formalization of previously self-employed workers, the mirror image of the transition observed in Events A and B.

These patterns are consistent with transitions from formal dependent employment to informal self-employment, but the cross-sectional design cannot trace individual workers. An alternative interpretation is that formal workers exit the labor force entirely while a compositionally different group of self-employed workers enters the affected zone independently. The 76% panel attrition documented in Appendix A precludes a definitive individual-level test. We interpret the self-employment evidence as suggestive of the transition channel, not proof of it.

5.2 Placebo Test

Applying the same estimator to artificial thresholds on the Event B population—fake increases from S/1,100→S/1,200 and S/1,400→S/1,500—produces ratios of 0.114 and 0.013, with missing mass 7 times smaller than at the true MW (1.2pp and 2.0pp vs. 8.0pp). The bunching signal is specific to the actual MW threshold.

Figure 2 shows the bunching ratio as a function of the excess window width $W \in \{50, 100, \ldots, 500\}$, holding missing mass fixed. $R(W)$ increases monotonically across all three events: from 0.62–0.79 at $W = \text{S}/50$ to 0.72–0.94 at $W = \text{S}/500$. Event C converges to $R = 0.94$ at $W = \text{S}/500$, consistent with workers dispersing broadly into bins above the new floor. Event A plateaus near $R = 0.72$, consistent with some genuine displacement into the informal or non-employment margin. For all events, the monotone increase indicates

that excess mass accumulates throughout the window rather than clustering in a narrow spike—ruling out pure wage-recording artifacts.

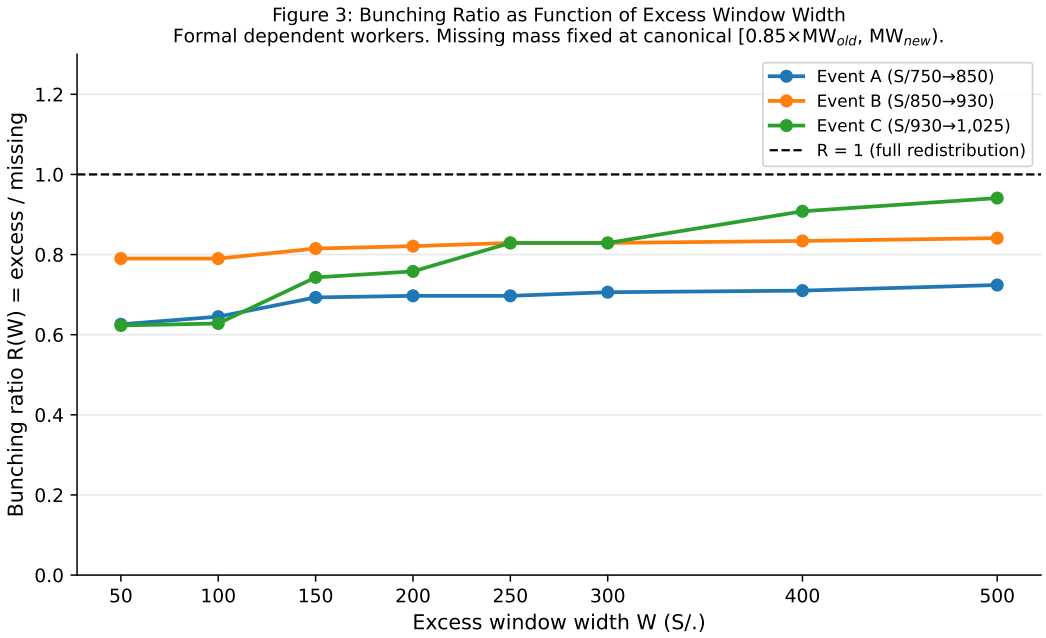

Figure 2: Bunching Ratio $R(W)$ as a Function of Excess Window Width

Note: Formal dependent workers (ENAHO CS). Missing mass fixed at canonical $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$. Excess window $[\text{MW}_{\text{new}}, \text{MW}_{\text{new}} + W)$ varies. Dashed line at $R = 1$. Event C: pre=2021, post=2023. Background correction: inverse-absolute weighted, clean zone $> 2 \times \text{MW}_{\text{new}}$.

Table 5 presents counterfactual sensitivity for all three events under six specifications. Canonical results are stable: alternative clean zone thresholds (1.5× and 2.5×) and no background correction all yield identical R across events, indicating background drift in the clean zone is near zero. Uniform weighting yields small downward adjustments (-0.035 to -0.056). The degree-4 polynomial yields $R = 0.456$, 1.054, and 2.085 for Events A, B, and C—increasingly unreliable as MW bite grows. Across the five non-polynomial specifications, the range of R is 0.638–0.697 (A), 0.773–0.829 (B), and 0.803–0.829 (C).

The near-zero background correction across all events reflects the stability of Peru's formal wage distribution above 2×MW during non-crisis periods. Aggregate formal employment

grew steadily at 2–4% per year over 2015–2019, with no differential growth in the upper versus lower tail. The near-zero drift is therefore genuinely reassuring rather than an artifact of offsetting biases. For Event C (2021–2023), the COVID recovery could have generated asymmetric growth in the upper tail, but the validation in Section 5 confirms the near-MW distribution was comparable in 2019 and 2021, and the background correction remains near zero.

The polynomial result warrants explanation. A degree-4 polynomial is fitted to the post-period wage distribution in the clean zone ($> 2 \times \text{MW}_{\text{new}}$) and extrapolated into the affected zone as the counterfactual level. Because the polynomial must pass through the peak of the distribution—which typically occurs above 1.5×MW—it extrapolates a high counterfactual into the missing zone, making the observed post-distribution appear to lie *above* the counterfactual there. This generates artificially small missing mass (the zone that "should" have workers according to the polynomial actually does) and consequently inflates R above 1. This is a well-known limitation of polynomial counterfactuals in bunching settings with sharp mass points (Kleven and Waseem, 2013): the polynomial cannot distinguish between the smooth background trend and the mass concentration created by the MW itself. The inverse-absolute weighting avoids this problem by assigning near-zero weight to bins near the MW spike and high weight to the flat upper tail, producing a counterfactual that reflects only smooth background drift.

Table 5: Counterfactual Sensitivity — Bunching Ratios, All Events

Specification	Clean zone / Weighting	R (Event A)	R (Event B)	R (Event C)
Canonical (inv-abs)	$> 2 \times$ MW, inv-abs	0.697	0.829	0.829
Uniform weights	$> 2 \times$ MW, uniform	0.638	0.773	0.803
Polynomial deg. 4	$> 2 \times$ MW, poly fit	0.456	1.054	2.085†
Clean at $1.5 \times$ MW	$> 1.5 \times$ MW, inv-abs	0.697	0.829	0.829
Clean at $2.5 \times$ MW	$> 2.5 \times$ MW, inv-abs	0.697	0.829	0.829
No correction	none	0.697	0.829	0.829

Note: Formal dependent workers. Missing zone: $[0.85 \times \mathrm{MW_{old}}, \mathrm{MW_{new}})$ (negative-only bins). Excess window: $[\mathrm{MW_{new}}, \mathrm{MW_{new}} + 250)$. Background correction varies by specification. Canonical and no-correction specifications give identical R because the clean-zone background is near zero for all three events.

† Polynomial overcorrection: the degree-4 fit extrapolates through the peak of the distribution, inflating the counterfactual in the missing zone. Event C's $R = 2.085$ reflects severe overcorrection at the largest MW bite (10.2%); see text.

Table 6: Bin Width Sensitivity — Bunching Ratios (Formal Dependent Workers)

Bin width	Boundary aligned? A / B / C	R (Event A)	R (Event B)	R (Event C)
S/10	Y / Y / N	0.751	0.836	0.850
S/15	N / Y / N	0.079†	0.838	0.848
S/20	N / N / N	0.712	0.804	0.854
S/25 (canonical)	Y / N / Y	0.697	0.829	0.829
S/50	Y / N / N	0.677	0.206†	0.662
S/100	N / N / N	0.477	0.754	0.613

Note: Canonical inv-abs background correction; excess window S/250; formal dependent workers. "Boundary aligned" = MW_{new} mod BinWidth = 0, so the MW threshold falls on a bin edge.

† Boundary misalignment: the bunching spike at MW_{new} falls in a bin whose center lies below MW_{new}, classifying it as missing rather than excess. See text.

The bin-boundary alignment requirement (Section 4.1) is confirmed in Table 6: at aligned configurations R is stable (range of 0.009 across S/10–S/25 for Event B; 0.006 for Event C), while severely misaligned widths produce catastrophic distortions ($R = 0.079$ for Event A at S/15; $R = 0.206$ for Event B at S/50). Event B at S/25 is marginally misaligned by S/5, which does not detectably distort R.

Event C and COVID. Event C's pre-period (2021) reflects post-pandemic conditions: aggregate formal employment fell 9% in 2020 and recovered partially by 2021, with some compression of wages near the MW floor. As a robustness check, using 2019 as the pre-period (a 4-year gap) produces $R = 0.997$, compared to $R = 0.830$ with the canonical 2021 pre-period. The pre-2019 estimate reflects upward bias from economy-wide wage growth over 2019–2023 (post-pandemic recovery, headline inflation of 33% cumulative), which shifts the entire distribution rightward and inflates measured excess mass. The canonical pre=2021 estimate is

temporally proximate (8 months before the May 2022 increase) and minimizes macro confounds; the higher pre-2019 R should be interpreted as an upper bound. Cumulative CPI inflation between 2019 and 2023 was approximately 20%. A naïve inflation adjustment of the 2019 distribution would shift the affected zone upward by ∼S/150, reducing overlap with the 2023 MW zone and inflating measured excess mass. The 2021 pre-period avoids this issue because cumulative inflation over 2021–2023 was only ∼8%, and the nominal MW (S/930) applies identically in both years. We therefore maintain the 2021 pre-period as canonical, with the 2019 estimate ($R = 0.997$) serving as an upper bound on the true ratio. Both estimates are consistent with the conclusion that employment effects cannot be signed. As a further validation, the formal dependent wage distribution in the affected zone $[791, 1{,}025)$ is nearly identical in 2019 (520,000 workers, 11.0% of all formal dep, median S/930) and 2021 (511,000 workers, 12.7%, median S/930), confirming that COVID did not compress the near-MW distribution of formal workers in a way that would bias the bunching estimator.

5.3 Wage Compression

Table 7 presents the wage compression DiD with a decomposition into mechanical (composition) and genuine components.

Table 7: Wage Compression DiD with Composition Decomposition

Event	Total DiD	Mechanical	Genuine	% Mech.	New arrivals
A (750→850)	−0.035	−0.028	−0.008	78%	13% of zone
B (850→930)	−0.050	−0.021	−0.029	41%	10% of zone
C (930→1,025)	−0.069	−0.064	−0.005	92%	26% of zone

Note: Compression zone = $[\text{MW}_{\text{new}}, 1.5 \times \text{MW}_{\text{new}}]$; high-wage zone = $[2 \times \text{MW}_{\text{new}}, 3 \times \text{MW}_{\text{new}}]$. Mechanical = predicted DiD from new arrivals entering at $\log(\text{MW}_{\text{new}})$ while existing workers grow at the high-wage rate. Genuine = total minus mechanical. With individual controls (age, sex, education, sector), the compression coefficient is -0.045 ($p < 0.001$).

For Event B, approximately 41% of the −0.050 DiD reflects the composition effect—workers pushed from below the MW to the new floor entering the compression zone and diluting its mean log wage. The remaining 59% (−0.029 log points) represents genuine relative wage compression. For Events A and C, the composition effect dominates (78% and 92%). The compression DiD is a descriptive statistic of distributional change, not a causal estimate. The distributional patterns we document are consistent with Engbom and Moser (2022), who find that Brazil's national MW compresses the wage distribution from below, and with Jales (2018), who uses a density discontinuity approach to estimate MW effects in Brazil.

Compression robustness. The canonical mechanical decomposition assumes new arrivals enter at exactly $\log(\text{MW}_{\text{new}})$ and existing compression-zone workers grow at the same rate as the high-wage control group. Table 8 relaxes both assumptions for Event B. Moving the assumed entry point from log(930) to log(980) or log(1,030) *reduces* the mechanical share (from 41% to 31% and 21%), increasing the genuine compression. This is because higher entry wages are less dilutive: new arrivals pull the mean down less, so more of the observed compression must reflect genuine suppression. Similarly, if existing workers grew at only

50% of the high-wage rate, the mechanical share rises marginally to 46%. In all four cases, genuine compression remains negative and economically meaningful (-0.027 to -0.040 log points), confirming that the compression result is not an artifact of the canonical entry-point assumption.

New arrivals as a share of the post-period compression zone vary substantially across events: 13% in Event A, 10% in Event B, and 26% in Event C (Table 7). Event C's higher share of new arrivals (26%) reflects both the larger absolute MW increase (S/95) and the post-COVID recovery in formal employment: the formal workforce grew 12.5% between 2021 and 2023, bringing a larger cohort of workers into formalization near the new floor of S/1,025. The high mechanical share in Event C (92%) is therefore an artifact of this post-pandemic compositional shift rather than evidence of weak genuine compression. For Event C, the upper-bound interpretation is particularly relevant. The mechanical decomposition assumes that new entrants' wages are determined independently of the MW—i.e., they enter at the floor because it is the market wage in their sector, not because the MW creates a mass point. If the MW increase drove formalization of previously informal workers (lighthouse channel), these entrants' wages are set by the MW itself, and classifying their compositional contribution as "mechanical" understates genuine compression. The 92% mechanical share should therefore be read as an upper bound on the non-MW-related component; the lower bound is the 8% genuine share from the canonical decomposition.

Table 8: Compression Robustness — Mechanical vs. Genuine (Event B)

Assumption	Mechanical	Genuine	% Mech.
1. Canonical: new arrivals at log(930)	−0.021	−0.030	41%
2. New arrivals at log(980)	−0.015	−0.035	31%
3. New arrivals at log(1,030)	−0.010	−0.040	21%
4. Existing workers grow at 50% of control rate	−0.023	−0.027	46%

Note: Event B (S/850→S/930, pre=2017, post=2019). Formal dependent workers. Total DiD fixed at −0.050 (observed). New arrivals = 10% of compression zone post-period ($n_{\text{new}}/n_{\text{post}} = 0.10$). Mechanical = predicted DiD contribution from composition change; genuine = total − mechanical. The high-wage growth rate $H = +0.005$ log points (from data). Entry points in assumptions 2–3 represent workers entering above the exact MW floor.

6 Heterogeneity

Organizing question. Do subgroups with higher MW exposure (lower wages, smaller firms, weaker enforcement) show systematically different bunching ratios? If enforcement drives compliance, we would expect larger firms and Lima (higher SUNAFIL presence) to show higher ratios. If market adjustment dominates, ratios should be similar across groups.

Table 9 through Table 11 present bunching ratios across subgroups. In all panels, the Total row (0.696 / 0.829 / 0.830) matches Table 3 exactly.

Table 9: Bunching Ratios by Industry Sector

Sector	Event A	Event B	Event C	Average
Total ($N = 9{,}952$)	**0.696**	**0.829**	**0.830**	**0.785**
Agriculture/Mining/Utilities ($N = 1{,}031$)	1.641	0.765	0.407	0.938
Manufacturing ($N = 904$)	1.000	0.869	1.049	0.973
Commerce ($N = 937$)	0.784	1.209	0.723	0.905
Transport/hospitality ($N = 556$)	1.041	1.018	0.585	0.881
Finance/professional services ($N = 1{,}113$)	0.768	1.018	0.806	0.864
Public administration ($N = 1{,}639$)	1.033	0.498	0.804	0.778
Education and health ($N = 3{,}021$)	0.598	0.709	0.975	0.760

Note: $N = 2015$ pre-period sample. Construction excluded (sparse cells, < 500 obs). Agriculture/Mining includes CIIU rev4 codes 35–39 (utilities). Sector from `p506r4` (CIIU 2-digit).

Table 10: Bunching Ratios by Firm Size, Age, and Sex

Group	Event A	Event B	Event C	Average
Panel B: Firm size				
Micro (≤ 10, $N = 705$)	0.899	0.909	0.690	0.833
Small (11–50, $N = 1{,}238$)	0.919	0.901	0.598	0.806
Medium/Large (51+, $N = 7{,}557$)	0.686	0.793	0.896	0.792
Panel C: Age and sex				
Age 18–24	0.661	1.010	0.805	0.826
Age 25–44	0.748	0.792	0.797	0.779
Age 45–64	0.701	0.927	0.853	0.827
Men	0.726	0.788	0.778	0.764
Women	0.697	0.922	0.877	0.832

Note: Firm size from `p512b` (establishment worker count). Workers with missing `p512b` (4.3–5.8% per year) are excluded from Panel B; the three rows sum to 95.5% of the sample.

Table 11: Bunching Ratios by Geography and Employer Sector

Group	Event A	Event B	Event C	Average
Lima Metropolitan ($N = 2{,}594$)	0.618	1.050	0.817	0.828
Rest of country ($N = 7{,}358$)	0.856	0.621	0.888	0.788
Urban ($N = 9{,}286$)	0.702	0.843	0.828	0.791
Rural ($N = 666$)	1.016	0.550	0.909	0.825
Public sector ($N = 4{,}315$)	0.921	0.456	0.865	0.747
Private sector ($N = 5{,}576$)	0.766	0.962	0.768	0.832

Note: Public sector identified by `p510` $\in \{1, 2, 3\}$ or CIIU $\in \{84, 85, 86\}$.

The evidence is mixed. The private sector absorbs MW increases more effectively than the

public sector (0.832 vs. 0.747), consistent with market adjustment rather than enforcement-driven compliance. Lima shows a higher ratio than the rest of the country in Event B (1.050 vs. 0.621), consistent with enforcement, but the pattern reverses in Events A and C. The absence of a consistent enforcement gradient suggests that formal-sector MW compliance in Peru operates through labor market adjustment—firms that choose to formalize have already internalized the MW floor—rather than through inspection-driven enforcement. Manufacturing consistently shows the highest ratios (≈ 0.97); micro firms show higher ratios than large firms in Events A–B, the opposite of an enforcement prediction, consistent with the self-selection story. No subgroup shows dramatically different behavior overall.

Interpretation caveat. The heterogeneity analysis is descriptive, not structural. There is no exogenous variation in enforcement intensity or sector-specific MW exposure to identify why ratios differ across subgroups. The firm-size pattern (micro > large in Events A–B) is inconsistent with a pure enforcement story and may reflect self-selection: micro firms that choose to formalize have already internalized compliance costs, so the marginal micro formal firm is already a high-compliance unit. The public/private gap (0.75 vs. 0.83) likely reflects institutional rigidity in government pay scales rather than differential MW responsiveness. These interpretations are suggestive, not causal.

Ethnicity. A natural question is whether MW increases differentially affect indigenous workers, who face higher barriers to formal employment. Building on Chávez (2024), who finds that MW increases benefit Andean workers in the lowest income brackets and help close the ethnic wage gap, the present analysis documents the underlying mechanism: the ethnic divide operates through differential access to formal employment (5.7% vs. 20.7% formality rate), not through differential wages within the formal sector. Among formal dependent workers, indigenous (Quechua/Aymara/native-language) and non-indigenous workers have similar wage distributions (both median S/1,400) and similar MW exposure (8.5% vs. 10.7% below MW). The 925 indigenous formal dependent workers are predominantly public-sector

employees (59%), concentrated in highland departments. Bunching ratios by ethnicity are unstable across events and reflect sector composition rather than differential MW compliance.

7 Hours

Table 12 presents the intensive-margin hours DiD.

Table 12: Weekly Hours DiD

Event	Treat pre	Treat post	Δ treat	Ctrl pre	Ctrl post	Δ ctrl	DiD
A (750→850)	44.4h	44.8h	+0.4h	42.1h	41.5h	−0.7h	+1.1h
B (850→930)	44.8h	45.1h	+0.3h	41.2h	42.0h	+0.8h	−0.5h
C (930→1,025)	44.5h	46.4h	+1.9h	42.1h	43.6h	+1.5h	+0.4h

Note: Treatment pre = workers in $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$; treatment post = workers in $[\mathrm{MW}_{\mathrm{new}}, 1.3 \times \mathrm{MW}_{\mathrm{new}}]$ (bunched-up zone). Control = $[1.5 \times \mathrm{MW}_{\mathrm{new}}, 3.0 \times \mathrm{MW}_{\mathrm{new}}]$. Weighted means using `fac500a`. Variable: `p513t` (total weekly hours).

No DiD exceeds ±1.1h/week. There is no evidence of intensive-margin hours adjustment in response to MW increases.

8 Conclusion

Peru's minimum wage increases between 2016 and 2022 produced clear, placebo-verified redistribution of the formal wage distribution. Across three events, 70–83% of workers displaced below the old floor reappear above the new one, rising to 94% when the excess window is extended to S/500 (Figure 2). The bunching signal is specific to the true MW threshold—7× larger than at placebo thresholds (Table 3)—stable across counterfactual specifications (Table 5), and confirmed on an independent quarterly dataset (EPE Lima, Appendix E). Hours show no intensive-margin adjustment ($\leq \pm 1.1$h/week). Wage compression of 3–7

percentage points is present, but 41–92% is mechanical composition; the residual genuine compression (-0.005 to -0.029 log points) is robust to alternative assumptions about where new arrivals enter (Table 8).

Employment effects remain an open question. The event study fails pre-trends for Events B ($p = 0.007$) and C ($p = 0.017$); Event A has no testable pre-period. The IV approach fails due to weak first stages ($F < 3$ in all events). The bunching ratios (0.70–0.83, bootstrap CIs [0.57–1.02]) are consistent with both modest displacement (up to 28% of affected workers) and full redistribution. The data cannot distinguish between these scenarios.

Three points of external validity strengthen these findings. First, EPE Lima quarterly data produces ratios (0.73–1.03) consistent with the ENAHO results despite different formality definitions, event windows, and geographic scope. Second, placebo tests at artificial thresholds produce ratios 7 times smaller. Third, informal workers show partial response in Event A (1.218) but not in Events B–C, suggesting some spillover at lower MW levels.

Policy implications. Peru's MW increases over 2016–2022 operated within a Kaitz range of 0.52–0.57, where the evidence supports effective redistribution: 70–83% of affected formal workers received raises, hours did not adjust, and the lighthouse evidence suggests some informal workers benefited as well. The cost was borne through institutional reallocation: some workers transitioned from formal dependent employment to informal self-employment, losing social security coverage while retaining employment and income at similar levels. Future MW adjustments that push the Kaitz index significantly above 0.60—as the 2025 increase to S/1,130 may do—enter territory where the balance between redistribution and displacement could shift, though our evidence cannot predict the threshold at which this occurs.

This paper's central contribution is a complete diagnostic of identification in a national-MW setting: the pre-post distributional estimator recovers a clear, placebo-verified signal

where three regression-based strategies—event study, Kaitz IV, and panel DiD—systematically fail. By documenting both what the data *can* and *cannot* identify, the paper provides a replicable template for MW analysis in developing economies without sub-national MW variation. This is not a data-quality failure—it is a structural feature of single-national-MW settings. The cross-sectional design cannot decompose the missing mass into transitions to informality versus exit from employment—a distinction with important policy implications. The self-employment evidence in Section 5 provides direct evidence that the missing mass reflects transitions to informal self-employment rather than job destruction; Corcuera (2024) corroborate this reallocation mechanism using an independent empirical strategy. Dube et al. (2010)'s border-discontinuity approach, which provides the cleanest identification in the U.S. context, is not feasible in Peru's national-MW setting—precisely the methodological tension this paper documents. If the 17–30% of the missing mass that does not reappear above the new floor reflects transitions to informal self-employment rather than non-employment—as the self-employment counts in Section 5 suggest—the welfare implications differ substantially: workers retain employment and income at similar levels while losing social security coverage. This extensive-margin informalization, rather than job destruction, is the likely channel through which Peru's MW increases impose costs on affected workers. Future work using administrative wage registries (SUNAT, ESSALUD) would allow this decomposition through quarterly firm-level job flows, substantially improving both the power to detect employment effects and the ability to trace where displaced workers go.

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A ENAHO Panel 978 — Event C

The ENAHO panel 978 tracks a subsample of individuals from 2021 to 2023. Treatment: formal dependent workers in 2021 with wages in [S/791, S/1,025). Control: wages in [S/1,230, S/2,563]. Approximately 76% of both groups are not re-interviewed; results apply to the surviving 24% only. Outcome statistics are weighted by `facpanel2123`. Attrited workers are *not* counted as non-employed.

Table 13: Panel Transition Results (Event C, 2021→2023)

Metric	Treatment ($N = 1{,}171$)	Control ($N = 3{,}903$)	DiD
Re-interviewed rate	24.4%	23.6%	—
Among re-interviewed (weighted):			
Employment retention	63.9%	60.1%	+3.8pp
Formal dep retention	46.8%	49.9%	−3.1pp
Δ log wage (stayers)	+0.307	+0.154	+0.153
Δ hours (stayers)	+0.7h	−1.3h	+2.0h

Caveat: 76% attrition with differential selection ($p = 0.022$). Do not interpret causally.

B Event Study — Pre-Trend Violations

All three events have identification problems. Event A: no prior years available (2015 = base); parallel trends untestable. Event B: joint pre-trend F-test $p = 0.007$; higher-Kaitz departments had systematically higher employment before the MW change. Event C: joint pre-trend $p = 0.017$; pre-COVID Kaitz–employment correlation renders the comparison non-parallel.

C IV / OWE — Not Identified

Three IV approaches were tested. (1) Department-level Kaitz IV: $F = 1.5/2.6/0.1$ across events, all far below the Stock-Yogo threshold. (2) Province-level IV: $F = 11.72$ for Event B but with reversed first-stage sign and failed pre-trends. (3) Within-year province IV: $F = 16.15$ but the placebo (no-MW-change year) produces $F = 12.71$—the instrument captures seasonal wage dynamics, not MW exposure.

Table 14: Department-Level Kaitz IV

Event	$\hat{\pi}$ (FS)	SE	F	$\hat{\beta}$ (RF)	SE	OWE	SE
A	+0.169	0.082	1.5	+0.014	0.038	+0.082	0.229
B	−0.070	0.103	2.6	+0.122	0.044	—	—
C	+0.132	0.090	0.1	+0.220	0.076	+1.667	1.271

Note: Event B OWE not reported (reversed first-stage sign). SE clustered by department (24 clusters). All $F < 3$; instrument is weak.

D EPEN CIU Annual 2023 — Lee-Saez Bunching

Single-period Lee-Saez estimate from INEI code 873 ($N = 53{,}316$ formal urban workers). All cities show excess mass at MW: ratio 1.622. These are not directly comparable to the pre-post ratios in Table 3.

E EPE Lima Quarterly Bunching

EPE Lima provides an independent dataset check with tighter 6-month event windows (Q1 vs. Q3 of each event year). Formality uses `p222 == 1` (EsSalud only), which is narrower than ENAHO. Lima Metropolitan Area only.

Table 15: EPE Lima Pre-Post Bunching (6-Month Windows)

Event	Window	N pre	N post	Near-MW	Ratio	95% CI	ENAHO
A	Q1→Q3 2016	2,622	2,599	240	1.031	[0.70, 1.63]	0.696
B	Q1→Q3 2018	2,280	2,438	302	0.733	[0.60, 1.03]	0.829
C	Q1→Q3 2022	1,933	1,936	331	0.885	[0.70, 1.13]	0.830

Note: All ENAHO ratios fall within EPE Lima bootstrap CIs. Event A diverges in point estimates; Lima's lower Kaitz may explain the higher absorption rate. CIs wide due to small N.